PRACTICAL NO:6

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def heapify(arr, n, i):
    largest = i
    left = 2 * i + 1
    right = 2 * i + 2
    if left < n and arr[left] > arr[largest]:
        largest = left
    if right < n and arr[right] > arr[largest]:
        largest = right
    if largest != i:
        arr[i], arr[largest] = arr[largest], arr[i]
        heapify(arr, n, largest)
def heapSort(arr):
    n = len(arr)
    # Build Max Heap
    for i in range(n // 2 - 1, -1, -1):
        heapify(arr, n, i)
    # Heap Sort
    for i in range(n - 1, 0, -1):
        arr[i], arr[0] = arr[0], arr[i]
        heapify(arr, i, 0)
# ---------------- Main ---------------- #
arr = []
n = int(input("Enter Number of Elements : "))
for i in range(n):
    x = int(input("Enter Element : "))
    arr.append(x)
print("\nOriginal Array")
```

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print(arr)
heapSort(arr)
print("\nSorted Array (Ascending)")
print(arr)
```